

SUBHRANIL DAS

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Experiences

Software Development Engineer

03/2024 - Present

School of Public Health, Indiana University

Remote, USA

- Developed software migrating large datasets (20+ **Excel, R**) from on-premise to **cloud storage**, using efficient data transfer, **API integrations** for enhanced engineering/data science accessibility.
- Designed, built, and automated scalable **ETL pipelines** using **Apache Airflow, AWS Lambda**; engineered for optimal performance, achieving **24% reduction** in data ingestion latency and increased data processing throughput across services.
- Engineered end-to-end data validation framework using custom **AWS Glue** jobs, integrated into **GitLab CI/CD pipelines**; system automated schema validation, cutting critical data processing errors in production by **15%**.
- Architected and implemented scalable cloud storage with **Amazon RDS, Amazon S3**, using intelligent partitioning and compression; this robust system managed a **30% data volume increase** while reducing monthly storage costs by **15%**.
- Developed real-time event ingestion/processing systems using **AWS Kinesis**, building anomaly detection capabilities and enabling **15% faster** insights via efficient stream processing.
- Implemented data governance protocols using software controls for data integrity, security (e.g., **AES-256 encryption**), and compliance (**GDPR, HIPAA**); this contributed to a **25% reduction** in data breach risk via robust system design.
- Collaborated with stakeholders to define requirements and develop real-time analytics; engineered interactive **Tableau dashboards** with optimized backend queries, **APIs**, leading to **23% improvement** in data-driven decision-making.
- Developed and maintained **RESTful APIs** for standardized, secure access to critical datasets, enabling seamless integration for internal applications and improving data discoverability and reusability.

Data Engineer

06/2021 - 07/2022

eGain Corporation

Pune, India

- Engineered high-performance SQL and Python ETL pipelines, effectively using **Azure Data Factory, Azure SQL Database** to elevate data flow reliability by **30%** across diverse industry sectors like healthcare.
- Designed and implemented scalable, highly optimized data models and architectural solutions within **Azure SQL Database**, directly enabling advanced analytics and fulfilling complex enterprise reporting for key stakeholders.
- Deployed proactive real-time monitoring and intelligent alerting platforms utilizing **Azure Monitor**, custom SQL triggers, markedly enhancing incident response efficiency, cutting critical issue resolution times by over **25%**.
- Developed insightful, interactive **Power BI dashboards** with advanced **DAX calculations**, seamlessly integrating datasets from **Azure Data Lake**, SQL sources, achieving a significant **35%** reduction in reporting lag times.
- Collaborated proactively and effectively within dynamic Agile/Scrum teams, leveraging **JIRA** for streamlined project management across 20+ high-impact deployments, consistently accelerating deployment velocity by **20%** per sprint.
- Systematically identified and thoroughly optimized critical SQL stored procedures, refined indexing strategies, and database partitioning on **Azure SQL**, achieving a notable **15%** acceleration in query retrieval times and improving dashboard responsiveness.
- Authored comprehensive, exceptionally user-friendly technical documentation for complex data pipelines, intricate **schemas, operational procedures**, fostering improved team knowledge sharing, expediting new engineer onboarding processes.

Skills

Programming/Scripting: Python, SQL, R, Java, C, C++

Databases: MySQL, PostgreSQL, MongoDB, Cassandra, Snowflake, NoSQL, PL/SQL, AWS Redshift, Spark/SQL, SSMS

Frameworks: Spark, Hadoop, PyTorch, TensorFlow, Keras, Django, Flask - RestAPI

Data Management and Analytics tools: dbT, Apache Airflow, Kafka, Power BI, Tableau, Excel, Google Analytics

CI/CD and DevOps: Kubernetes, Docker, Linux, JIRA, Git, GitLab, Jenkins, Bitbucket

Cloud Platforms: AWS - S3, Glue, Sagemaker, Kinesis, EMR, Athena, Lambda, Google Cloud Platform - Bucket, BigQuery

Projects

Dynamic-Commentary | *Python, YouTube API, Pydub, ChatGPT API, NLTK, Data Visualization*

[Link](#)

- Directed a project on tonal shifts and narrative strategies in esports commentary, enhancing stakeholder insights, and improving sentiment analysis accuracy by **20%** using **ChatGPT API** and **NLTK's VADER**.
- Created interactive **Python** visualizations to map sentiment plots, identifying key tonal shift moments of the commentators accompanied with that of the live audience chat with **90%** precision.

Turbocharge Retail Insights | *Python, Apache Airflow, dbT, Soda, Docker, GCP, Big Query, Metabase*

[Link](#)

- Oversaw the development of a comprehensive **Apache Airflow** ETL pipeline integrating **GCP Bucket, BigQuery, Soda**, and **dbt**, achieving **25%** enhancement in process efficiency.
- Optimized over **20** SQL scripts and dbt models, and implemented automated testing procedures (**DAGs**) for financial data ingestion, quality checks, and transformations, significantly improving data quality.

Generative AI for Pathology Datasets | *Python, Tensorflow, Scikit-learn, Keras, OpenCV*

[Link](#)

- Led a project on **Generative AI** for pathology datasets, utilizing **GANs** to synthesize nuclear detection datasets in medical imaging while addressing **HIPAA**-related data access restrictions to create synthetic data for model training.
- Utilized advanced GAN architectures such as **DCGAN, Variational Autoencoders**, and **StyleGAN3**, enhancing model performance by **40%**, and trained a **YOLOv8** architecture to achieve an impressive accuracy of **85%** in nuclei detection models.

Education

Indiana University, Bloomington

08/2022 - 05/2024

Master of Science in Data Science

Coursework: Machine Learning, Data Mining, Advanced Database Concepts, Data Visualization, Applied Algorithms, Deep Learning Systems, Applied Database Technologies, Statistics & Information Retrieval.